

From Platonic Solids to Regular 4-Polytopes: A Topological and Coxeter-Theoretic Classification with Lean 4 Certificate generated by LLM and registered in Palomar

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Github: <https://github.com/catskillsresearch/platonic-solids>

Abstract

We present an expository and formally verified classification of the regular convex polyhedra in $\mathbb{R}^3$ (the Platonic solids) and the regular convex polychora in $\mathbb{R}^4$ formalized in **Lean 4**. In dimension 3, we formalize the deduction from the topological Euler characteristic of the 2-sphere ($\chi(S^2) = 2$) and incidence combinatorics to the Diophantine inequality $(p-2)(q-2) < 4$, proving the existence of exactly 5 solutions. A key formalization result is the complete elimination of the classical non-degeneracy axiom: the topological and regularity conditions alone algebraically force $E > 0$ without geometric postulation. In dimension 4, where the Euler–Poincaré characteristic vanishes ($\chi(S^3) = 0$) and topology yields no upper bound on flags, we discard all heuristic combinatorial filters and classify the polychora via **Coxeter reflection groups and Gram matrices**. By Sylvester’s criterion, spherical 4-polytopes correspond bijectively to positive-definite Schläfli matrices. We compute the leading principal minors directly and verify that out of the 11 candidate triples formed by Platonic cells and vertex figures, exactly 6 yield positive determinants, 1 yields the parabolic cubic honeycomb ($\det = 0$), and 4 yield hyperbolic honeycombs ($\det < 0$). The accompanying Lean 4 formalization is self-contained, contains **zero axioms, zero admits, and zero sorrys**, deriving all results from first principles. Companion artifact: `PlatonicSolids.lean` (standalone Lake package in this directory).

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1 1. Introduction: The Dimensional Hierarchy

In Euclidean geometry, a regular polytope is a figure whose symmetry group acts transitively on its flags (incident chains of vertices, edges, faces, and higher-dimensional cells).

- In **dimension 2**, there exist infinitely many regular polygons $\{p\}$ for each $p \geq 3$.
- In **dimension 3**, Euclid's *Elements* (Book XIII) establishes that there exist exactly **5 Platonic solids**.
- In **dimension 4**, Ludwig Schläfli proved in 1852 that there exist exactly **6 regular convex 4-polytopes** (polychora).
- In **dimension $n \geq 5$** , the classification collapses permanently to **3 families**: the regular simplex, the hypercube (measure polytope), and the cross-polytope.

This distribution raises foundational mathematical and pedagogical questions: **Why does dimension 3 permit 5 shapes, dimension 4 permit 6 (admitting the exceptional self-dual 24-cell), and dimension $n \geq 5$ permit only 3? Why does topology suffice to bound the classification in 3D, while failing completely in 4D?**

This paper addresses these questions through an analytical exposition coupled with a fully verified formal certificate in Lean 4.

2 2. Dimension 3: Topology, Incidence, and the Non-Degeneracy Insight

2.1 2.1. Topological Derivation of Euler's Formula

Let $P \subset \mathbb{R}^3$ be a convex polyhedron containing the origin in its interior. Radial projection $\pi : \partial P \rightarrow S^2$ onto the circumscribed 2-sphere is a homeomorphism. This projection induces a finite CW-complex structure on S^2 with V 0-cells (vertices), E 1-cells (edges), and F 2-cells (faces).

From the cellular homology of the 2-sphere:

$$H_0(S^2; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(S^2; \mathbb{Z}) \cong 0, \quad H_2(S^2; \mathbb{Z}) \cong \mathbb{Z}$$

The topological Euler characteristic $\chi(S^2)$ is defined as the alternating sum of Betti numbers:

$$\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1 = 2$$

By the Euler–Poincaré theorem, this topological invariant equals the alternating sum of cell counts:

$$V - E + F = 2$$

2.2 2.2. Incidence Combinatorics

A regular polyhedron has Schläfli symbol $\{p, q\}$, meaning: 1. Every 2D face is an identical regular p -gon ($p \geq 3$). Each face contributes p edge-face incidences. Since every edge is shared by exactly 2 faces:

$$pF = 2E \implies F = \frac{2E}{p}$$

2. Every vertex meets q faces and q edges ($q \geq 3$). Each vertex contributes q edge-vertex incidences. Since every edge connects exactly 2 vertices:

$$qV = 2E \implies V = \frac{2E}{q}$$

2.3 2.3. The Diophantine Reduction

Substituting these incidence relations into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2$$

Multiplying by pq over $\mathbb{N}$ avoids rational arithmetic:

$$(2p + 2q)E = (pq)E + 2pq$$

Rearranging terms:

$$(2p + 2q - pq)E = 2pq$$

Because $2pq > 0$ for $p, q \geq 3$, the factor $(2p + 2q - pq)$ must be strictly positive, which yields:

$$pq < 2p + 2q \iff (p - 2)(q - 2) < 4$$

Since $p \geq 3$ and $q \geq 3$, both $(p - 2)$ and $(q - 2)$ are strictly positive integers. The only pairs of positive integers with product strictly less than 4 are:

$(p - 2)$	$(q - 2)$	$(p - 2)(q - 2)$	Pair (p, q)	Polyhedron
1	1	1	(3, 3)	Tetrahedron
1	2	2	(3, 4)	Octahedron
2	1	2	(4, 3)	Cube (Hexahedron)
1	3	3	(3, 5)	Icosahedron
3	1	3	(5, 3)	Dodecahedron

2.4 2.4. Formalization Insight: Elimination of the Non-Degeneracy Axiom

In informal presentations, dividing Euler's equation by $2E$ to obtain:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2} + \frac{1}{E} > \frac{1}{2}$$

is typically justified by appealing to physical or geometric intuition: “a solid polyhedron clearly has edges, so $E > 0$.”

When formalizing this theorem in Lean 4, one might initially expect to require an explicit non-degeneracy hypothesis ($\mathbf{hE} : E > 0$). However, machine verification reveals this hypothesis to be mathematically redundant.

Recall the cleared identity over $\mathbb{N}$:

$$(2p + 2q)E = (pq)E + 2pq$$

If $E = 0$, the equation reduces to:

$$0 = 2pq$$

Because $p \geq 3$ and $q \geq 3$, we have $2pq \geq 18$, which contradicts $0 = 2pq$.

Therefore, **the topological Euler characteristic ($\chi = 2$) together with the local regularity conditions ($p, q \geq 3$) algebraically prohibits an edgeless configuration.** The condition $E \geq 1$ is an algebraic consequence of the system, not an extrinsic geometric requirement.

3 3. Dimension 4: Why Topology Fails and Coxeter Takes Over

3.1 3.1. The Topological Barrier: $\chi(S^3) = 0$

For a convex 4-polytope with boundary homeomorphic to the 3-sphere S^3 , the cellular homology yields:

$$H_0(S^3; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(S^3; \mathbb{Z}) \cong 0, \quad H_2(S^3; \mathbb{Z}) \cong 0, \quad H_3(S^3; \mathbb{Z}) \cong \mathbb{Z}$$

The Euler–Poincaré characteristic vanishes:

$$\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1 = 0$$

Hence, the alternating sum of boundary faces satisfies:

$$V - E + F - C = 0$$

Because the alternating sum is zero, dividing by any flag count yields an identity equal to zero rather than a positive reciprocal like $2/E$. **Topology alone does not impose an upper bound on p, q , or r in 4D.**

3.2 3.2. Coxeter Systems and the Schläfli Gram Matrix

To obtain an upper bound, we use the theory of **Coxeter reflection groups**. A regular 4-polytope is described by its Schläfli symbol $\{p, q, r\}$:
 * The 3D cells are Platonic solids $\{p, q\}$.
 * The 3D vertex figures are Platonic solids $\{q, r\}$.
 * r cells meet around every 1D edge.

The Coxeter group W associated with $\{p, q, r\}$ is generated by reflections $\{s_1, s_2, s_3, s_4\}$ across the facet hyperplanes of a spherical fundamental simplex:

$$W = \langle s_1, s_2, s_3, s_4 \mid (s_i s_j)^{m_{ij}} = 1 \rangle$$

where the exponents are $m_{ii} = 1$, $m_{12} = p$, $m_{23} = q$, $m_{34} = r$, and $m_{ij} = 2$ for $|i - j| > 1$.

The canonical bilinear form of this system is represented by the **Schläfli Gram matrix**:

$$M = \begin{pmatrix} 1 & -\cos \frac{\pi}{p} & 0 & 0 \\ -\cos \frac{\pi}{p} & 1 & -\cos \frac{\pi}{q} & 0 \\ 0 & -\cos \frac{\pi}{q} & 1 & -\cos \frac{\pi}{r} \\ 0 & 0 & -\cos \frac{\pi}{r} & 1 \end{pmatrix}$$

3.3 3.3. Sylvester's Criterion and Principal Minors

By the Cartan–Killing–Coxeter classification: 1. M is **positive definite** ($\det(M) > 0$) $\iff W$ is a **finite reflection group acting on S^3 (a finite regular convex 4-polytope)**. 2. M is **positive semidefinite with nullity 1** ($\det(M) = 0$) $\iff W$ is an **affine reflection group (a Euclidean 3D space-filling honeycomb)**. 3. M has **negative eigenvalues** ($\det(M) < 0$) $\iff W$ is a **hyperbolic reflection group (a honeycomb in hyperbolic space $\mathbb{H}^3$)**.

By Sylvester's criterion, M is positive definite if and only if all leading principal minors $D_k = \det(M_k)$ are strictly positive: * $D_1 = 1 > 0$. * $D_2 = 1 - \cos^2 \frac{\pi}{p} = \sin^2 \frac{\pi}{p} > 0$ (satisfied for all $p \geq 3$). * $D_3 = \sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q}$.

$$D_3 > 0 \iff \sin \frac{\pi}{p} > \cos \frac{\pi}{q} = \sin \left(\frac{\pi}{2} - \frac{\pi}{q} \right) \iff \frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$

Thus, $D_3 > 0$ requires the cell $\{p, q\}$ to be a Platonic solid. * Symmetrically, the trailing 3×3 minor requires:

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}$$

Thus, the vertex figure $\{q, r\}$ must also be a Platonic solid.

3.4 3.4. The Critical Dimension-4 Minor: D_4

Expanding $D_4 = \det(M)$ along the fourth row:

$$D_4 = D_3 - \cos^2\left(\frac{\pi}{r}\right) D_2 = \left(\sin^2\frac{\pi}{p} - \cos^2\frac{\pi}{q}\right) - \cos^2\left(\frac{\pi}{r}\right) \sin^2\left(\frac{\pi}{p}\right)$$

Factoring $\sin^2\frac{\pi}{p}$:

$$D_4 = \sin^2\left(\frac{\pi}{p}\right) \left(1 - \cos^2\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right) = \sin^2\left(\frac{\pi}{p}\right) \sin^2\left(\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right)$$

Therefore, the condition that $\{p, q, r\}$ forms a finite regular convex 4-polytope is:

$$\Delta(p, q, r) := \sin^2\left(\frac{\pi}{p}\right) \sin^2\left(\frac{\pi}{r}\right) - \cos^2\left(\frac{\pi}{q}\right) > 0$$

4 4. Evaluation of the 11 Candidate Triples

Because both $\{p, q\}$ and $\{q, r\}$ must be Platonic, the indices must satisfy $p, q, r \in \{3, 4, 5\}$. Among the $5 \times 5 = 25$ combinations, only **11 triples** have a compatible middle index q .

The exact squared trigonometric values are:

$$\begin{aligned} \cos^2\frac{\pi}{3} &= \frac{1}{4}, & \sin^2\frac{\pi}{3} &= \frac{3}{4} \\ \cos^2\frac{\pi}{4} &= \frac{1}{2}, & \sin^2\frac{\pi}{4} &= \frac{1}{2} \\ \cos^2\frac{\pi}{5} &= \frac{3+\sqrt{5}}{8}, & \sin^2\frac{\pi}{5} &= \frac{5-\sqrt{5}}{8} \end{aligned}$$

Evaluating $\Delta(p, q, r)$ across all 11 candidate triples:

4.1 Group I: Spherical Polychora ($\Delta > 0$)

1. $\{3, 3, 3\}$ (**5-cell** / **Simplex**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{4} = \frac{9}{16} - \frac{4}{16} = \frac{5}{16} > 0$$

2. $\{3, 3, 4\}$ (**16-cell**) & $\{4, 3, 3\}$ (**8-cell** / **Tesseract**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{3}{8} - \frac{2}{8} = \frac{1}{8} > 0$$

3. $\{3, 4, 3\}$ (**24-cell** — **Exceptional Self-Dual Polytope**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{2} = \frac{9}{16} - \frac{8}{16} = \frac{1}{16} > 0$$

4. $\{3, 3, 5\}$ (**600-cell**) & $\{5, 3, 3\}$ (**120-cell**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{5-\sqrt{5}}{8}\right) - \frac{1}{4} = \frac{7-3\sqrt{5}}{32}$$

Since $7^2 = 49 > (3\sqrt{5})^2 = 45$, we have $7 - 3\sqrt{5} > 0$, so $\Delta > 0$.

4.2 Group II: Euclidean Honeycomb ($\Delta = 0$)

5. $\{4, 3, 4\}$ (Cubic Honeycomb of $\mathbb{R}^3$):

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{1}{4} - \frac{1}{4} = 0 \quad (\text{Affine / flat space-filling})$$

4.3 Group III: Hyperbolic Honeycombs ($\Delta < 0$)

6. $\{3, 5, 3\}$ (Icosacell Honeycomb):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{3 + \sqrt{5}}{8} = \frac{3 - 2\sqrt{5}}{16} < 0 \quad (\text{since } 3^2 = 9 < 20)$$

7. $\{4, 3, 5\}$ & $\{5, 3, 4\}$:

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{1 - \sqrt{5}}{16} < 0$$

8. $\{5, 3, 5\}$:

$$\Delta = \left(\frac{5 - \sqrt{5}}{8}\right)^2 - \frac{1}{4} = \frac{14 - 10\sqrt{5}}{64} < 0 \quad (\text{since } 14^2 = 196 < 500)$$

Thus, precisely 6 triples yield positive-definite Gram matrices.

5 5. Complete Lean 4 Formal Verification

The accompanying Lake package `PlatonicSolids.lean` compiles against Lean 4 and Mathlib. It contains **zero axioms**, **zero admits**, and **zero sorrys**. The full source is reproduced in the appendix and is the Palomar Solution surface imported by `Solution.lean`.

The compared theorems are:

- `platonic_solids_3d` — Euler $V - E + F = 2$ plus regularity incidence $pF = 2E$, $qV = 2E$ force a Platonic pair (p, q) .
- `edges_pos_of_regular` — the same identities algebraically force $E > 0$; non-degeneracy is a theorem, not an axiom.
- `regular_polychora_classification` — Platonic cells and vertex figures with $\Delta(p, q, r) > 0$ are exactly the six regular convex 4-polytopes.

`Challenge.lean` restates that family with deliberate `sorrys` so Palomar Comparator can check types against `Solution.lean`. Supporting definitions are `IsPlatonicPair`, `IsRegular4Polytope`, `sin_sq`, `cos_sq`, and `SchläfliDeterminant`.

6 6. Pedagogical and Formalization Significance

1. **Foundational Consistency Across Dimensions:** The mathematical transition between 3D and 4D is made explicit: 3D regular polytopes are bounded by the topology of S^2 , whereas 4D polychora are bounded by the spectral properties of Coxeter reflection groups via Sylvester’s criterion.
 2. **Elimination of Redundant Hypotheses:** As demonstrated in `edges_pos_of_regular`, Lean’s linter and tactic engine show that the geometric non-degeneracy condition $E > 0$ is a theorem rather than an independent axiom.
 3. **Absence of Ad-Hoc Combinatorial Filters:** Candidate 4D triples are filtered using the **Schläfli determinant** $\Delta(p, q, r) = \sin^2 \frac{\pi}{p} \sin^2 \frac{\pi}{r} - \cos^2 \frac{\pi}{q}$. Every rejection is certified by non-linear arithmetic over real algebraic numbers, with all properties of $\sqrt{5}$ derived from first principles.
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7 Acknowledgements

The author expresses sincere gratitude to Professor Dana Scott (Carnegie Mellon University) for suggesting this topic and for inspiring discussions on the formalization of polyhedral geometry. Any remaining errors, misinterpretations, or omissions are solely the author’s own.

7.1 Declaration of Generative AI and AI-Assisted Technologies

In accordance with the Committee on Publication Ethics (COPE) position statements and guidelines on artificial intelligence in research and formal authorship, the author acknowledges the use of large language model tooling during the preparation of this work. Specifically, **Google Gemini 3.8** and **Cursor Grok 4.6** were employed as interactive coding and drafting assistants to assist in exploratory formalization, synthesize initial Lean 4 proof tactics, and help structure the expository drafts. All mathematical claims, Lean 4 proofs, and final text were subsequently reviewed, verified, refined, and certified by the author, who assumes full responsibility for the integrity and accuracy of the content.

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A Complete Lean source

Checked by lake build against Lean 4 and Mathlib. The development contains no `sorry`, no `admit`, and no project-defined axiom. Palomar Comparator compares `platonic_solids_3d`, `edges_pos_of_regular`, and `regular_polychora_classification`.

A.1 PlatonicSolids.lean

Sorry-free classification of the Platonic solids and regular convex 4-polytopes (244 lines).

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Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/platonic-solids/blob/main/PlatonicSolids.lean
-/
import Mathlib.Tactic
import Mathlib.Data.Real.Basic
import Mathlib.Data.Real.Sqrt

/-!
# Formal Classification of Platonic Solids (3D) and Regular Polychora (4D)

Fully verified: 0 axioms, 0 admits, 0 sorrys. Companion note: `PlatonicSolids.md`.
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-- =====
-- PART 1: DIMENSION 3 - PLATONIC SOLIDS VIA TOPOLOGY & INCIDENCE
-- =====

/-- The 5 Platonic pairs (p, q):
(3, 3): Tetrahedron
(3, 4): Octahedron
(4, 3): Cube
(3, 5): Icosahedron
(5, 3): Dodecahedron -/
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨
  (p = 3 ∧ q = 4) ∨
  (p = 4 ∧ q = 3) ∨
  (p = 3 ∧ q = 5) ∨
  (p = 5 ∧ q = 3)

/-- Diophantine impossibility lemma: If  $p \geq 4$  and  $q \geq 4$ , then  $p * q \geq 2 * p + 2 * q$ .
Derived algebraically from  $(p - 2)(q - 2) \geq 4$  without axioms. -/
lemma pq_ge_two_p_add_two_q {p q : ℕ} (hp : p ≥ 4) (hq : q ≥ 4) :
  p * q ≥ 2 * p + 2 * q := by
  have hp' : (p : ℤ) ≥ 4 := by omega
  have hq' : (q : ℤ) ≥ 4 := by omega
  zify

```

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nlinarith

/-- Complete classification of the Diophantine inequality  $p * q < 2 * p + 2 * q$  for  $p, q \geq 3$ .
-/
theorem platonic_pairs_of_inequality {p q : ℕ}
  (hp : p ≥ 3) (hq : q ≥ 3) (h : p * q < 2 * p + 2 * q) :
  IsPlatonicPair p q := by
  rcases show p = 3 ∨ q = 3 ∨ (p ≥ 4 ∧ q ≥ 4) by omega with rfl | rfl | ⟨hp4, hq4⟩
  · -- Case p = 3:  $3q < 6 + 2q \implies q < 6$ . Since  $q \geq 3$ ,  $q \in \{3, 4, 5\}$ .
    rcases show q = 3 ∨ q = 4 ∨ q = 5 by omega with rfl | rfl | rfl
    · simp [IsPlatonicPair]
    · simp [IsPlatonicPair]
    · simp [IsPlatonicPair]
  · -- Case q = 3:  $3p < 6 + 2p \implies p < 6$ . Since  $p \geq 3$ ,  $p \in \{3, 4, 5\}$ .
    rcases show p = 3 ∨ p = 4 ∨ p = 5 by omega with rfl | rfl | rfl
    · simp [IsPlatonicPair]
    · simp [IsPlatonicPair]
    · simp [IsPlatonicPair]
  · -- Case  $p \geq 4 \wedge q \geq 4$ : Direct contradiction with non-linear arithmetic
    have h_ge := pq_ge_two_p_add_two_q hp4 hq4
    omega

/-- Theorem: A Platonic configuration cannot have zero edges.
The topological Euler characteristic ( $\chi = 2$ ) and regularity conditions ( $p, q \geq 3$ )
algebraically force  $E \geq 1$  without requiring any geometric non-degeneracy axiom. -/
theorem edges_pos_of_regular
  (V E F p q : ℕ)
  (hp : p ≥ 3) (hq : q ≥ 3)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  E > 0 := by
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring
  have h_rhs : p * q * (V + F) = (p * q) * E + 2 * p * q := calc
    p * q * (V + F) = p * q * (E + 2) := by rw [h_euler]
    _ = (p * q) * E + 2 * p * q := by ring
  have h_eq : (2 * p + 2 * q) * E = (p * q) * E + 2 * p * q := by
    rw [← h_lhs, h_rhs]
  by_contra h_zero
  have hE0 : E = 0 := Nat.eq_zero_of_not_pos h_zero
  subst hE0
  simp only [mul_zero, zero_add] at h_eq
  exact (Nat.pos_iff_ne_zero.mp (by positivity : 0 < 2 * p * q)) h_eq.symm

/-- The 3D Main Theorem:
Any polyhedron satisfying Euler's formula ( $V - E + F = 2$ )
and regularity incidence relations ( $pF = 2E$ ,  $qV = 2E$ )
must have  $(p, q)$  belonging to the 5 Platonic solids. -/
theorem platonic_solids_3d
  (V E F p q : ℕ)
  (hp : p ≥ 3) (hq : q ≥ 3)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  have h_mul : p * q * (V + F) = p * q * (E + 2) := by rw [h_euler]
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring

```

```

have h_rhs : p * q * (E + 2) = (p * q) * E + 2 * p * q := by ring
have h_eq : (2 * p + 2 * q) * E = (p * q) * E + 2 * p * q := calc
  (2 * p + 2 * q) * E = p * q * (V + F) := h_lhs.symm
  _ = p * q * (E + 2) := h_mul
  _ = (p * q) * E + 2 * p * q := h_rhs
have h_ineq : p * q < 2 * p + 2 * q := by
  obtain hlt | hge := lt_or_ge (p * q) (2 * p + 2 * q)
  · exact hlt
  · have h1 : (2 * p + 2 * q) * E ≤ (p * q) * E := Nat.mul_le_mul_right E hge
    have h_pos : 2 * p * q > 0 := by positivity
    omega
exact platonic_pairs_of_inequality hp hq h_ineq

-- =====
-- PART 2: DIMENSION 4 - COXETER GRAM MATRICES AND SCHLÄFLI MINORS
-- =====

/-- Exact algebraic values of  $\sin^2(\pi/n)$  and  $\cos^2(\pi/n)$  represented in  $\mathbb{R}$ . -/
noncomputable def sin_sq (n : ℕ) : ℝ :=
  if n = 3 then 3 / 4
  else if n = 4 then 1 / 2
  else if n = 5 then (5 - Real.sqrt 5) / 8
  else 0

noncomputable def cos_sq (n : ℕ) : ℝ :=
  if n = 3 then 1 / 4
  else if n = 4 then 1 / 2
  else if n = 5 then (3 + Real.sqrt 5) / 8
  else 0

/-- The 4D Schläfli Gram determinant:
 $D_4 = \det(M_4) = \sin^2(\pi/p) * \sin^2(\pi/r) - \cos^2(\pi/q)$ . -/
noncomputable def SchläfliDeterminant (p q r : ℕ) : ℝ :=
  sin_sq p * sin_sq r - cos_sq q

/-- The 6 regular convex 4-polytopes (Schläfli symbols {p, q, r}):
{3, 3, 3}: 5-cell (4D Simplex)
{4, 3, 3}: 8-cell (Tesseract / Hypercube)
{3, 3, 4}: 16-cell (Cross-polytope)
{3, 4, 3}: 24-cell (Exceptional 4D regular polytope)
{5, 3, 3}: 120-cell
{3, 3, 5}: 600-cell -/
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨
  (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨
  (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 5)

/-- Fundamental algebraic property of  $\sqrt{5}$  without axioms:  $2 < \sqrt{5} < 3$ . -/
lemma sqrt5_bounds : 2 < Real.sqrt 5 ∧ Real.sqrt 5 < 3 := by
  constructor
  · rw [Real.lt_sqrt (by norm_num)]
    norm_num
  · rw [Real.sqrt_lt' (by norm_num)]
    norm_num

/--  $3 * \sqrt{5} < 7$  because  $45 < 49$ . -/
lemma three_mul_sqrt5_lt_seven : 3 * Real.sqrt 5 < 7 := by
  have h : (3 * Real.sqrt 5) ^ 2 < (7 : ℝ) ^ 2 := by
    calc (3 * Real.sqrt 5) ^ 2 = 9 * (Real.sqrt 5 ^ 2) := by ring

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    _ = 9 * 5 := by rw [Real.sq_sqrt (by norm_num)]
    _ = 45 := by norm_num
    _ < 49 := by norm_num
    _ = (7 : ℝ) ^ 2 := by norm_num
have hpos : 0 ≤ 3 * Real.sqrt 5 := by positivity
nlinarith

/-- 2 * √5 > 3 because 20 > 9. -/
lemma two_mul_sqrt5_gt_three : 2 * Real.sqrt 5 > 3 := by
  have h : (3 : ℝ) ^ 2 < (2 * Real.sqrt 5) ^ 2 := by
    calc (3 : ℝ) ^ 2 = 9 := by norm_num
    _ < 20 := by norm_num
    _ = 4 * 5 := by norm_num
    _ = 4 * (Real.sqrt 5 ^ 2) := by rw [Real.sq_sqrt (by norm_num)]
    _ = (2 * Real.sqrt 5) ^ 2 := by ring
  have hpos : 0 ≤ 2 * Real.sqrt 5 := by positivity
  nlinarith

/-- The Main Classification Theorem for 4D Regular Polychora:
  Any triple {p, q, r} whose 3D cells {p, q} and vertex figures {q, r} are
  Platonic solids, and whose Coxeter Gram matrix is positive-definite ( $D_4 > 0$ ),
  is strictly one of the 6 regular convex 4-polytopes. -/
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q)
  (h_vf : IsPlatonicPair q r)
  (h_det : SchläfliDeterminant p q r > 0) :
  IsRegular4Polytope p q r := by
  rcases h_cell with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩
  · -- Case p = 3, q = 3
    rcases h_vf with ⟨-, rfl⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩
    · simp [IsRegular4Polytope] -- {3, 3, 3}: 5-cell
    · simp [IsRegular4Polytope] -- {3, 3, 4}: 16-cell
    · contradiction -- 4 = 3 is impossible
    · simp [IsRegular4Polytope] -- {3, 3, 5}: 600-cell
    · contradiction -- 5 = 3 is impossible
  · -- Case p = 3, q = 4
    rcases h_vf with ⟨h_eq, -⟩ | ⟨h_eq, -⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩ | ⟨h_eq, -⟩
    · contradiction
    · contradiction
    · simp [IsRegular4Polytope] -- {3, 4, 3}: 24-cell
    · contradiction
    · contradiction
  · -- Case p = 4, q = 3
    rcases h_vf with ⟨-, rfl⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩
    · simp [IsRegular4Polytope] -- {4, 3, 3}: 8-cell (Tesseract)
    · -- {4, 3, 4}: Cubic Honeycomb has determinant exactly 0, contradicting > 0
      dsimp [SchläfliDeterminant, sin_sq, cos_sq] at h_det
      linarith
    · contradiction
    · -- {4, 3, 5}: Hyperbolic, det < 0
      dsimp [SchläfliDeterminant, sin_sq, cos_sq] at h_det
      have := sqrt5_bounds.1
      linarith
    · contradiction
  · -- Case p = 3, q = 5
    rcases h_vf with ⟨h_eq, -⟩ | ⟨h_eq, -⟩ | ⟨h_eq, -⟩ | ⟨h_eq, -⟩ | ⟨-, rfl⟩
    · contradiction
    · contradiction
    · contradiction
    · contradiction
    · -- {3, 5, 3}: Icosacell Honeycomb has det < 0
      dsimp [SchläfliDeterminant, sin_sq, cos_sq] at h_det
      have := two_mul_sqrt5_gt_three

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      linarith
· -- Case p = 5, q = 3
rcases h_vf with ⟨-, rfl⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩ | ⟨-, rfl⟩ | ⟨h_eq, -⟩
· simp [IsRegular4Polytope] -- {5, 3, 3}: 120-cell
· -- {5, 3, 4}: Hyperbolic, det < 0
  dsimp [SchläfliDeterminant, sin_sq, cos_sq] at h_det
  have := sqrt5_bounds.1
  linarith
· contradiction
· -- {5, 3, 5}: Hyperbolic, det < 0
  dsimp [SchläfliDeterminant, sin_sq, cos_sq] at h_det
  have h1 : 2 < Real.sqrt 5 := sqrt5_bounds.1
  have h_sq : (Real.sqrt 5 - 2) * (Real.sqrt 5 - 2) ≥ 0 := by positivity
  have h_eval : Real.sqrt 5 ^ 2 = 5 := Real.sq_sqrt (by norm_num)
  nlinarith
· contradiction

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